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RDC Testing with Webcam Eye Tracking

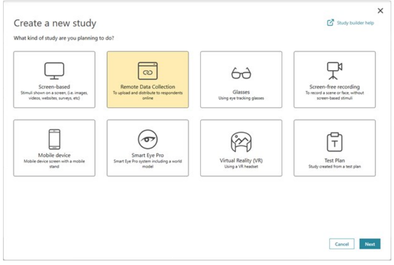
This article walks through the steps for creating an iMotions study for Remote Data Collection with Web Camera-based eye tracking.

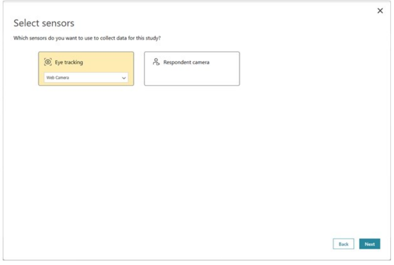
Study Requirements

- Eye Tracking must be enabled with *Web Camera* selected as the eye tracker.
- Pre-study and Post-study Calibration slides must be included (included by default for Remote Data Collection studies).

Enabling webcam eye tracking

1. Create a Remote Data Collection study
2. Enable Eye tracking
3. Select Web Camera

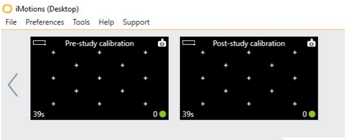

Create a new study
Study builder help
What kind of study are you planning to do?
Screen-based
Stimuli shown on a screen, (i.e. images, videos, websites, surveys, etc)
Remote Data Collection
To upload and distribute to respondents online
Glasses
Using eye tracking glasses
Screen-free recording
To record a scene or face, without screen-based stimuli
Mobile device
Mobile device screen with a mobile stand
Smart Eye Pro
Smart Eye Pro system including a world model
Virtual Reality (VR)
Using a VR headset
Test Plan
Study created from a test plan
Cancel
Next


Select sensors
Which sensors do you want to use to collect data for this study?
Eye tracking
Respondent camera
Back
Next

Eye tracking: Web Camera

Calibration slides

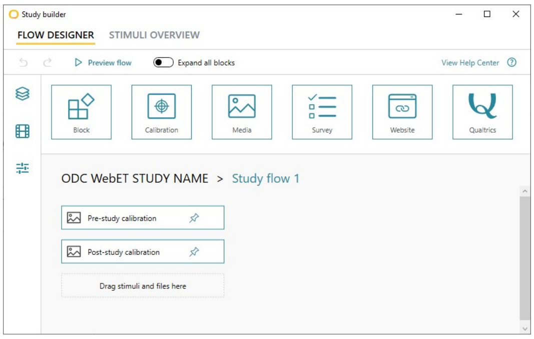
Calibration slides are automatically included at the start and end of remote data collection studies where Eye tracking is selected as a sensor (in iMotions 10 or newer). A 13-point calibration is shown, where respondents should maintain fixation on each calibration point as they are shown.



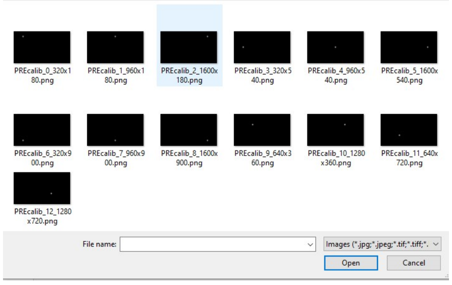
Adding calibration slides to your study

Additional calibration slides can be added to your study to improve eye tracking. Additional calibration is generally recommended for every 10 minutes of study stimuli.

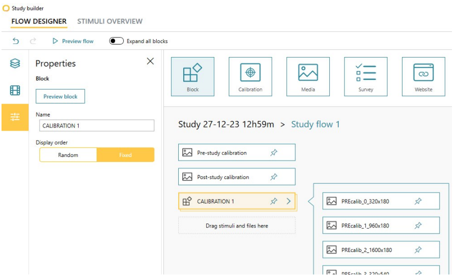
1.) To begin adding calibration slides to your new study, click "Edit stimuli" to bring up the Study Builder interface.



2.) Click Calibration and navigate to any folder containing Webcam Eye Tracking calibration slides and select one (or more) image files, such as those provided by iMotions.



3.) After calibration image files have been selected, they will appear as a "CALIBRATION" block.



Name: CALIBRATION 1

Customizing Calibration Stimuli

Custom calibration images, coordinates, and durations can be used for calibration stimuli.

Unique naming convention of calibration slides

For your RDC study, your calibration slides must follow a particular naming convention in order for the platform to identify that a specific image stimulus should be utilized as part of the calibration of the webcam eye tracking algorithms.

Plaintext Copy

```
[name]_[index]_[x-pos]x[y-pos].png
```

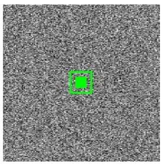
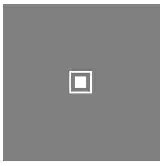
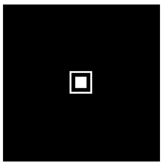
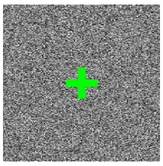
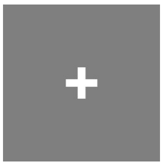
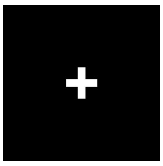
[name] is a user defined string, like `calibration` or `calib`. Do not use special characters or underscore (_) in the name.
[index] is a integer value. Must be the same value for different calibration images with same (x,y) position, but with different name.
[x-pos] is x position (in pixel) of the calibration mark in relation to image resolution. Top-Left is (0,0).
[y-pos] is y position (in pixel) of the calibration mark in relation to image resolution.

Note: Best practice is to have the resolution of your calibration images be the same resolution as your study resolution! You can use either a JPG or PNG image for your calibration slides.

Sample calibration slides

We have included six (6) sets of high-resolution (1920x1080) calibration images for you to use. In addition to these calibration slides, you can always create your own calibration images using the instructions above.

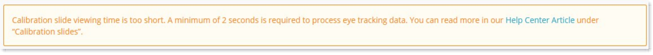
Each calibration set provided below contains 13 images, denoting 13 different calibration locations on the screen (i.e., 13-point calibration). If you decide that you need a shorter calibration procedure, then you can use the first 9 images in each set for a 9-point calibration.

High Contrast Background	Gray Background	Black Background
	 Calib_Gray_Squares.zip	 Calib_Black_Squares.zip
	 Calib_Gray_Crosshairs.zip	 Calib_Black_Crosshairs.zip

Calibration slide duration

For each calibration slide, the viewing time should be set for 2 seconds at minimum. We recommend 4 seconds per calibration slide as the exposure time. Please remember, you can only set this exposure time when you are creating your study in the desktop software prior to uploading it to the cloud for data collection.

Note: If your calibration slide is < 2 seconds, you will receive the warning shown below (on the online dashboard) and you will not be able to access the "Start remote data collection" button. In this scenario, you will need to delete your study from the cloud system and re-upload it with the correct calibration timing.



Calibration slide viewing time is too short. A minimum of 2 seconds is required to process eye tracking data. You can read more in our Help Center Article under "Calibration slides".

RECOMMENDATION: Have at least two sets of calibration, once at the start of your study and once more at the end of your study, using at least 9 calibration points each time with 4 seconds of exposure per calibration slide.

Animated calibration slides

It is also possible to present an animated calibration screen to the participant, in which the calibration dots move to each position. A short example of how this appears is shown in the gif below:

This is added to an RDC study by adding the following files for post- and pre-study calibration, respectively:

- WebcamCalibrationSlideTemplates\Animated_Calibration_Placeholders\Post-study calibration.png
- WebcamCalibrationSlideTemplates\Animated_Calibration_Placeholders\Pre-study calibration.png

The folder that these are located in ("Animated_Calibration_Placeholders") will appear when you select to add calibration slides from the desktop as stimuli. When the calibration slides are later viewed in iMotions Lab, a single image with all 13 fixation points will be shown.

Sample study template

For a sample study set up for webcam-based eye tracking, please download RDC_SampleStudy_wET_TwoCalibrations.zip (also available at the bottom of the article as an attachment). In this study, participants are asked to:

1. Go through an initial 13-pt calibration (square targets on gray background)
2. Watch one video ad & examine two print ads
3. Go through a second 13-pt calibration (square targets on gray background)

Prior to each step above, written instructions are also provided to inform the participant what they are about to undergo. This sample study uses the same set of calibration slides provided in the Calibration_SquareTarget_Gray.zip file below. For this study, we utilized two calibration steps: one in the beginning of the study and one at the end of the study, bookending the study contents. This is currently the best practice that we recommend for webcam eye tracking.

Note: This sample study was created using iMotions build 8.2.5. As such, you will need this specific build of iMotions, or newer, to access this sample study.

Note 2: This sample study is meant to be a simple (using only video and image stimuli) to showcase the calibration set up. However, you can also set up your webcam-based eye tracking study to also test websites and include questionnaires using external survey platforms, such as Qualtrics.

WebET Whitepaper

iMotions WebET 3.0 whitepaper